M U H A M M A D H A M Z A

Ph.D. Candidate, Peking University | CFD / FSI Simulation & Embedded Sensing

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PROFESSIONAL SUMMARY

Ph.D. candidate at Peking University (expected 2027) working at the intersection of fluid-structure simulation and embedded sensing, with 3+ years of prior industry R&D experience as a CFD/CAE engineer in aerospace, renewable energy, and fluid machinery. On the simulation side, specialized in aerodynamic and hydrodynamic shape optimization, FSI and FEM analysis, and surrogate-based design optimization. On the sensing side, develops embedded piezoelectric (PVDF) sensing for soft bodies in flow, from signal-conditioning hardware and synchronized data acquisition to machine-learning estimation of body shape, flow conditions, and position, deployed in real time in water-channel tests. 8 papers (2 first-author, 6 published) in venues including IEEE/ASME Transactions on Mechatronics, Elsevier Energy, and ICRA. Awarded a competitive 200,000 RMB NSFC research grant. Proficient in ANSYS Fluent, ANSYS ICEM, Star-CCM+, COMSOL Multiphysics, OpenFOAM, IBAMR, Femap/Nastran, MATLAB, and Python.

EDUCATION

Ph.D. in General Mechanics and Foundation of Mechanics Sep 2023 – Jul 2027 (expected)
Peking University (北京大学), Beijing, China
Chinese Government Scholarship recipient
Supervisor: Xie Guangming

- Research focus: Bio-inspired soft bodies in flow — energy harvesting, proprioception (self-deformation sensing), upstream body localization, and external flow/environment sensing via PVDF piezoelectric signals
- Combine CFD optimization, FSI simulation, and experimental water-channel validation with machine-learning interpretation of sensor signals
- Developed multi-physics COMSOL models coupling fluid dynamics, structural mechanics, and piezoelectric response
- First-author papers in Elsevier Energy and ICRA (under review), plus 6 co-authored papers including IEEE/ASME Transactions on Mechatronics, Advanced Intelligent Systems, and Proceedings of the Royal Society B

M.S. in Mechanics Sep 2019 – Jul 2021
Xi'an Jiaotong University (西安交通大学), Xi'an, China
Chinese Government Scholarship recipient
Supervisor: Zhang Yang

- Thesis: Discrete adjoint method for aerodynamic shape optimization — applied to the ONERA M6 wing and NREL S809 vortex generators (ANSYS Fluent, ANSYS ICEM), achieving improved L/D through constrained and unconstrained optimization
- Conducted RANS simulation and grid-convergence study on the DLR F6 wing-body configuration, validated against established experimental data

B.S. in Aerospace Engineering Oct 2014 – Jul 2018
Institute of Space Technology, Islamabad, Pakistan
Supervisor: Izhar Hussain Kazmi

- Thesis: Design and analysis of a tube-launched micro aerial vehicle (MAV) with foldable wings, including thrust-to-weight (T/W) and wing-loading (W/S) optimization, pneumatic launcher design, and stability augmentation system design

INTERNATIONAL RESEARCH EXPERIENCE

Visiting Researcher Oct – Dec 2024
Max Planck Institute of Animal Behavior, Konstanz, Germany

- **3D fish-body CFD (IBAMR):** Ran immersed-boundary CFD simulations of 3D fish bodies in IBAMR and conducted grid independence studies to establish mesh-converged solutions
- **In-code timestep interpolation:** Re-engineered the IBAMR kinematics handling to interpolate prescribed body velocities to finer timesteps directly within the solver — removing the prior dependence on externally pre-written velocity files and enabling arbitrary timestep refinement without regenerating input data
- **Data pipeline optimization:** Wrote surface-probe extraction routines for the fish geometry; reduced point-cloud and probe-data read time by up to 60% and output file size by up to 70%

WORK EXPERIENCE

CAE Engineer, R&D Oct 2022 – Sep 2023
TenFong (Zhejiang) Technology Co., Ltd., China

- **Multi-domain CFD simulation:** Conducted aerodynamic and hydrodynamic analyses (Star-CCM+, ANSYS Fluent) across wind energy, water pump, and urban development projects
- **Water pump FSI optimization:** Performed fluid-structure interaction analysis on water pump assemblies in Star-CCM+, identifying geometry modifications that reduced internal pressure losses

- **Multiphase casting simulation:** Modeled molten-iron furnace pouring as a transient two-phase (liquid–air) free-surface flow using the interFoam VOF solver in OpenFOAM, capturing tank-filling and air displacement behavior

Aero Engineer, R&D Aug 2021 – Sep 2022
Wanhu Aerospace Industry Co., Ltd., China

- **Aircraft CFD analysis:** Performed full-aircraft RANS simulations for a FAR-23 category general aviation aircraft (ANSYS ICEM, Fluent, CFD-Post), generating aerodynamic data for flight envelope assessment
- **Composite & loads analysis:** Conducted composite layup stress analysis and flight loads analysis using Femap/Nastran to support FAR-23 certification requirements
- **Design iteration support:** Integrated CFD results with structural constraints to inform iterative airframe design decisions across the development cycle

Aero Engineer Aug 2018 – Aug 2019
H&A Bots (Pvt.) Ltd., Islamabad, Pakistan

- **Simulation-driven design:** Performed CFD and structural simulations for conceptual design of aerospace products, evaluating aerodynamic and hydrodynamic performance across operating conditions
- **Performance modeling:** Developed 1D system performance models in MATLAB to support early-stage trade studies and design decisions

TECHNICAL SKILLS

CFD / CAE: ANSYS Fluent, ANSYS ICEM CFD, Star-CCM+, COMSOL Multiphysics (FSI, Laminar, Turbulent SST), OpenFOAM (interFoam), IBAMR
FEA / Structural: Femap/Nastran, ANSYS Static Structural, composite stress analysis, loads analysis
Numerical Methods: FVM, FEM, immersed boundary, ALE moving mesh, VOF multiphase, RANS / SST k-ω turbulence modeling
Optimization & Surrogate Modeling: Discrete adjoint, genetic algorithm (GA), Nelder-Mead, screening methods; surrogate models — Kriging, neural networks, polynomial response surface (2nd-order), non-parametric regression (NPR)
Programming: MATLAB (primary), Python (acquisition automation), C++ (IBAMR solver modification), LaTeX
Machine Learning: Sequence models (GRU, LSTM, BiLSTM), kernel SVR with random-feature approximation, ARX system identification, surrogate modeling for design optimization, time-series regression, temporal convolutional networks (TCN), real-time causal inference on streaming sensor data, state estimation and localization from distributed sensor signals
Post-processing: Tecplot 360, ParaView, CFD-Post
Sensing & Experimental: PVDF piezoelectric sensor integration, signal-conditioning front ends (OPA4134 unity-gain buffers, opto-isolated triggering), synchronized multi-sensor acquisition (NI-DAQmx, Basler cameras), subsonic water channel (Lolligo), subsonic wind tunnel, NI LabVIEW DAQ (USB-6210, myRIO-1900)
CAD / Prototyping: SolidWorks, CATIA V5, FDM 3D printing (slicing, calibration), rapid prototyping

SELECTED PROJECTS

- **Underwater Wake Sensing and Localization —** Extended the PVDF soft-body platform to estimate upstream wake scale and the body's streamwise and lateral position relative to an upstream bluff body from embedded PVDF signals in a water channel. Benchmarked six estimators (linear, kernel, random forest, MLP, LSTM, TCN), evaluated generalization to unseen conditions, flow speeds, and measurement sessions, and identified the best sensor placement through ablation. Demonstrated real-time estimation in live water-channel tests (≈36 ms per update), tracking flow-speed changes and following streamwise repositioning within about 4 s.
- **PVDF Soft-Foil Proprioception —** Built a multi-channel PVDF soft-body platform with a servo-actuated excitation rig for dynamic midline shape reconstruction of a soft foil in flow. Designed a 4-channel OPA4134 unity-gain buffer front end that eliminates multiplexer crosstalk between high-impedance PVDF channels on the single-ADC NI USB-6210; automated hardware-synchronized acquisition in Python (NI-DAQmx + Basler camera) capturing exact frame counts at exact timestamps aligned to the PVDF voltage stream; opto-isolated the camera trigger (PC817) to keep mains noise and ground loops out of the DAQ. Benchmarked five reconstruction models in MATLAB (kernel SVR, ARX, GRU, LSTM, BiLSTM); the BiLSTM reconstructed the dynamic midline to 0.69 mm RMSE (R^2 = 0.83) on test flow conditions, with trailing-edge error at 0.62% of body length. Submitted to ICRA.
- **Hydrofoil Energy Harvester Optimization —** Built a parametric CFD + surrogate-modeling framework (two-way COMSOL FSI) to optimize a bio-inspired flexible hydrofoil for flow-energy harvesting; surrogate-driven optimization improved harvested power by up to 122% over the baseline foil. Fabricated and water-channel-tested designs delivered up to 30% higher measured electrical output, with successful field deployment in a natural stream. Published in Elsevier Energy.
- **Discrete-Adjoint Aerodynamic Shape Optimization —** Built a gradient-based shape-optimization workflow with the ANSYS Fluent discrete adjoint solver and mesh morphing on structured ICEM meshes ($y+ \approx 0.9$), validated against wind-tunnel data. ONERA M6 transonic wing (Mach 0.84, Re 12 million, k-ε): drag reduced by 16.0% and L/D increased by 20.7% unconstrained, and drag reduced by 9.6% with L/D up 13.3% under an upper-surface-only shape constraint. Counter-rotating vortex generators on the NREL S809 wind-turbine airfoil (SST k-ω, 5.6 M-cell mesh from a six-grid independence study): at the 14° stall angle, drag reduced by 29.5% and L/D increased by 49.9% (25.1 → 37.7).
- **Tube-Launched MAV with Foldable Wings —** Designed a miniature air vehicle that stows in a 127-mm (5-inch) launch tube and deploys foldable wings and twin tails. Sized it with a T/W–W/S study over nine design points, which showed T/W should rise from 0.59 to 1.3, matched motor, battery and propeller, and defined

the flight envelope (load factors +5/−2). Designed the pneumatic tube launcher (4.6 bar for a ~20 m/s exit speed at a 30° launch angle), a roll controller and a lateral-directional stability augmentation system in MATLAB/Simulink using Tornado VLM stability derivatives, and analyzed the external flow in ANSYS Fluent from a CATIA model.

PUBLICATIONS

8 papers (2 first-author): 6 published, 2 under review. Venues include *IEEE/ASME Transactions on Mechatronics*, Elsevier *Energy*, *Advanced Intelligent Systems*, *Proceedings of the Royal Society B*, ICRA, and *IEEE Robotics and Automation Letters*.

- "An Optimization Framework for Soft Fluidic Actuators: Fully Coupled FSI Simulation with Surrogate Modeling." IEEE/ASME Transactions on Mechatronics, 2026. doi:10.1109/TMECH.2026.3748317
  Ahsan Tanveer, Waqar Hussain Afridi, Rahdar Hussain Afridi, **Muhammad Hamza**, Mingxin Wu, Ke Liu, Guangming Xie
- "Toward Gripper-Integrated Active Electrosense for Pre-Contact Sensing in Underwater Soft Grippers." Workshop on Manipulation Robustness, ICRA 2026. doi:10.48550/arXiv.2606.03204
  Ahsan Tanveer, **Muhammad Hamza**, Waqar Hussain Afridi, Chen Wang, Guangming Xie
- "A Geometric Optimization Framework for a Bio-inspired Flexible Hydrofoil Energy Harvester." Energy (Elsevier), 2026. doi:10.1016/j.energy.2026.140825
  Muhammad Hamza, Ahsan Tanveer, Waqar Hussain Afridi, Rahdar Hussain Afridi, Li Jie, Chen Wang, Guangming Xie
- "EMG-Driven Telemetry and Inference System for Fish: Pose Reconstruction and Flow Sensing." Advanced Intelligent Systems, 2026. doi:10.1002/aisy.202501085
  Rahdar Hussain Afridi, Waqar Hussain Afridi, **Muhammad Hamza**, Ahsan Tanveer, Mingxin Wu, Xingwen Zheng, Liang Li, Guangming Xie
- "Bio-to-Robot Transfer of Fish Sensorimotor Dynamics via Interpretable Model." Advanced Intelligent Systems, 2025. doi:10.1002/aisy.202501117
  Waqar Hussain Afridi, Ahsan Tanveer, Rahdar Hussain Afridi, **Muhammad Hamza**, Mingxin Wu, Liang Li, Guangming Xie
- "Beyond Propulsion: Muscle Proprioception Enables Hydrodynamic Sensing in Fish Body." Proceedings of the Royal Society B, 2025. doi:10.1098/rspb.2025.0474
  Rahdar Hussain Afridi, Waqar Hussain Afridi, **Muhammad Hamza**, Mingxin Wu, Li-Ming Chao, Yufan Zhai, Liang Li, Guangming Xie
- "Explicit Actuator-Effectiveness Estimation for Post-Training Robot Policy Recovery." IEEE Robotics and Automation Letters (under review).
  Ahsan Tanveer, **Muhammad Hamza**, Rahdar Hussain Afridi, Waqar Hussain Afridi, Chen Wang, Zongqing Lu, Guangming Xie

GRANTS & SCHOLARSHIPS

- **NSFC RFEIPS Grant (2024):** 200,000 RMB — National Natural Science Foundation of China Research Fund for Excellent International PhD Students (competitively awarded)
- **Chinese Government Scholarship — Ph.D.:** Full funding, Peking University, 2023–2027
- **Chinese Government Scholarship — Master's:** Full funding, Xi'an Jiaotong University, 2019–2021

HONORS & CERTIFICATIONS

- Social Work Award (社会工作奖) — Peking University, 2026 (university-level honor)
- Machine Learning Certificate — Stanford University
- 63rd IKCEST Silk Road Training Programme on Artificial Intelligence
- 60th IKCEST Silk Road Training Programme on Aerospace
- "Excellent Student" Award, XJTU Silk Road Winter Camp
- Oral Presenter, Student Research Paper Conference (SRPC), Institute of Space Technology
- Best Performance Award, Aeromodelling Competition